EARTHPRESS · PROJECT PLAN

EarthPress project plan

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/earthpress		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-25	Amish Chadha	First project plan (kit 1.8 pipeline) at stage P3, with the reference list for the repo

EarthPress project plan

EarthPress is at stage P3 (design outputs, TRL 3): the lever and toggle press is calculated, modelled at detail level D3 with its two alternative concepts, drawn, priced and planned. The next gate, P4 (verification, TRL 4), is on hold under the portfolio cap. Decisions requested from Amish: the concept selection (D1, three options within 10 %, no pick) and the R7 mass total (D2, recommendation: set it at 205 kg). Method: STANDARDS sections 19 and 23.

Workstreams

Table 1. Workstreams.

Workstream	Owner (role)	Deliverable	Status
Concept and STEM check	Design lead	EPR-STM-001, EPR-OPT-001, EPR-DDR-004	Done at P3; STEM check open on soil data (S2, S6, S18); D1 awaiting Amish
Requirements and traceability	Design lead	EPR-URS-001, EPR-PRS-001, EPR-RTM-001	Done at P3; to validate with users
Calculation and design	Design lead	EPR-CAL-001, `cad/src/model.py`, drawings EPR-DWG-001 and 101 to 115	Done at P3 (fix F1 applied); D2 awaiting Amish
Build plan and BOM	Design lead	EPR-BLD-001, `bom/bom.csv`	Done at P3
Regulatory, risk, market	Design lead	EPR-REG-001, EPR-RSK-001, EPR-MKT-001	First pass done; standards marked "to assess"
Co-design partner	Amish	Partner agreement, co-design sessions	Not started (candidate decided 2026-10-02)
Renders and storefront	Coordinator on Amish's Mac	Options picture, product renders, cards	Waiting for the Mac render
Verification (P4)	Fabricator and partner	Prototype build, soil compaction test, block tests, proof load	On hold (portfolio cap P3)

Resources needed

Table 2. Resources (P4 onward is on hold; estimates are indicative).

Resource	Deliverable	Effort	Estimated cost (USD)
Design lead	P3 upkeep, decisions D1 and D2 carried through	1 to 2 days	Not costed
Local welding shop (P4, on hold)	One prototype press to EPR-BLD-001	About 5 shop days	542 parts plus labor (labor not estimated)
Co-design partner (P4, on hold)	Soil compaction curves, timed cycles, user feedback	2 to 4 weeks	To assess
Testing laboratory (P4, on hold)	Block compressive and wet strength to PR-003	1 batch	To assess

Stage gates

Table 3. Stages (P4 to P6 on hold while the stage cap is P3).

Stage	Gate criteria	Status
P0 Proposal	PRB, STM, OPT, REG, FEA, PLN	Met (2026-10-25)
P1 Definition	URS, PRS, RSK with hazard analysis, MKT	Met (2026-10-25)
P2 Design inputs	PRS with targets and verification, CAL, RTM, concept DDR, STEM and feasibility closed	Met except the STEM check, open on soil data (repo kept at TRL 3 under STANDARDS section 20) and the concept decision D1
P3 Design outputs	Model, STEP, drawings, priced BOM, BLD, DEC, design FMEA, traceability complete	Met (2026-10-25); CAD fidelity D3 for every option
P4 Verification	Prototype, test plans and reports, verification report	On hold
P5 Validation	Field trials with a partner	On hold
P6 Commercialization	Only with a named deployment partner	On hold

Next steps and decision requested

1. Amish: decide D1 (concept: A, B or C; all within 10 %, no pick) and D2 (R7 mass total; recommendation: 205 kg including bushes, nuts, washers and weld metal).
2. Coordinator: render the options picture and the product renders on the Mac (`python .kit/options.py` , `.kit/export_views.py` , `.kit/photoreal.py` , `.kit/cards.py`).
3. When the cap allows TRL 4: approach the partner, measure the soil compaction curve (closes STM S6), build the prototype, and test blocks (closes S2, S18).

References

Every source cited in this repo's documents.

- Auroville Earth Institute. "Compressed Stabilised Earth Block." Accessed 2026-10-25. https://www.earth-auroville.com/compressed_stabilised_earth_block_en.php.
- Auroville Earth Institute. "Specifications of the Press 3000." Accessed 2026-10-25. https://www.earth-auroville.com/specifications_of_the_press_3000_en.php.
- Botti. Article on the CINVA-Ram and self-help housing. *Frontiers of Architectural Research*, 2023. https://www.sciencedirect.com/science/article/pii/S2095263523000584 (as cited in the README; title not rechecked).
- Budynas, Richard G., and J. Keith Nisbett. *Shigley's Mechanical Engineering Design*. New York: McGraw-Hill.
- Bureau of Indian Standards. IS 1725:2013 and IS 1725:2023, *Stabilized Soil Blocks Used in General Building Construction: Specification*. Accessed 2026-10-25. https://www.services.bis.gov.in/php/BIS_2.0/bisconnect/knowyourstandards/Indian_standards/isdetails/NDQ4.
- Chatham House. *Making Concrete Change: Innovation in Low-Carbon Cement and Concrete*. 2018. https://www.chathamhouse.org/2018/06/making-concrete-change-innovation-low-carbon-cement-and-concrete.

- Construction Kenya. Soil block press article. 2024-01-23. Accessed 2026-10-25. https://www.constructionkenya.com/?p=4970.
- Engineering for Change. "Makiga Stabilized Soil Block Press." Updated 2024-03-04. Accessed 2026-10-25. https://www.engineeringforchange.org/solutions/product/makiga-stabilized-soil-block-press/.
- Global ABC Sustainable Materials Hub. "African Standard WD-ARS 1333:2018 Compressed Stabilized Earth Blocks." Accessed 2026-10-25. https://globalabc.org/sustainable-materials-hub/resources/african-standard-wd-ars-13332018-compressed-stabilized-earth.
- IARC. *Arsenic, Metals, Fibres, and Dusts*. IARC Monographs 100C. Lyon: IARC, 2012.
- Natural Building Blog. Post on compressed earth block presses. Accessed 2026-10-25. https://naturalbuildingblog.com/?p=2586.
- Neville, A. M. *Properties of Concrete*. Harlow: Pearson.
- New Mexico Commission of Public Records. 14.7.4 NMAC, *Earthen Building Materials Code*. Accessed 2026-10-25. https://www.srca.nm.gov/parts/title14/14.007.0004.html.
- Open Source Ecology. "CEB Press." https://wiki.opensourceecology.org/wiki/CEB_Press.
- OSHA. "Silica, Crystalline: Construction." Accessed 2026-10-25. https://osha.gov/silica-crystalline/construction-info.
- Rwanda Standards Board. DRS 514-1:2022, *Compressed Earth Blocks, Part 1*. Accessed 2026-10-25. https://www.rsb.gov.rw/fileadmin/user_upload/files/pictures/DRS_514-1_2022.pdf.
- Swiss Steel Group. "42CrMoS4" data sheet. Accessed 2026-10-25. https://swisssteel-group.com/content-media/documents/Data-Sheets/Engineering-Steel/42CrMoS4_en.pdf.
- UNEP. *Global Status Report for Buildings and Construction 2024/2025*. https://www.unep.org/resources/report/global-status-report-buildings-and-construction-20242025.
- UN-Habitat. "Housing." https://unhabitat.org/topic/housing.
- United Nations. *The Sustainable Development Goals Report 2024*, Goal 11. https://unstats.un.org/sdgs/report/2024/Goal-11/.
- Standards cited by number and title: ASTM D698; ASTM E2392/E2392M-24; EN 1993-1-9; ISO 11228-1; ISO 11228-2; ISO 12100; ISO 13857; Regulation (EU) 2023/1230; 29 CFR 1926.1153.